

Alejandro Hernandez Bautista

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Education

B.Eng. in Computer Engineering

Jan 2020 – Jun 2025

Major: Computer Systems and Software Engineering

Minor: Artificial Intelligence and Intelligent Systems

Coursework: Mathematics (1-3) | Physics | Statistics | Data Structures (1-4) | Algorithms (1-2) | Compilers and Interpreters (1-4) | Programming (1-3) | Web Programming (1-2) | Databases (1-2) | Artificial Intelligence (1-4) | Electronics and Circuits (1-2) | Software Engineering (1-2) | Computer Architecture (1-2)

Professional Experience

Software Startup (SaaS)

May 2026 – Present

Co-Founder & Backend Developer

Developed and deployed the backend infrastructure for an HR ERP SaaS application, designing core application services and data architecture to support business operations. Managed application deployment, version control workflows, and production maintenance, including troubleshooting and hotfixes for early business clients (backend development, database architecture, cloud deployment, production support).

City Science Lab GDL | MIT Media Lab

Aug 2024 - Feb 2026

Data Analyst & UI/UX Researcher

Sep 2025 - Feb 2026

Conducted data analysis for a low-cost water management system, applying machine learning techniques to filter sensor noise and identify usage patterns that compensated for hardware limitations. Contributed to UI/UX research to evaluate user interaction and system accessibility (data analysis, machine learning, backend, research).

Backend Developer

Feb 2025 - Aug 2025

Developed and maintained APIs for a low-cost water management and sensing system to support communication between IoT devices, databases, and user interfaces, enabling reliable data collection, storage, and visualization through full-stack web application (database architecture, backend, frontend).

Embedded Systems Developer

Aug 2024 - Jan 2025

Developed embedded software for ESP32-based devices in a distributed low-cost water management and sensing system, implementing communication protocols and networking features. Worked on device integration, testing, and deployment in informal communities and university campuses (embedded programming, electronics, IoT, backend).

Continental Automotive

Sep 2024 - Sep 2025

Automation & Digitization Developer Intern

Developed full-stack web applications in smart manufacturing environments, enabling process automation and digitization. Worked on system deployment, the implementation of robotic solutions, and user training for digital tools and automation platforms, and delivered technical demonstrations and workshops for students from different universities (automation, robotics, backend, frontend, deployment, NoSQL database, embedded programming).

Projects

Distributed Water Monitoring System (Axol)

City Science Lab Guadalajara

Integrated IoT devices, databases, and user interfaces to monitor water quality and consumption in informal communities; applied machine learning to improve sensor reliability under hardware constraints (electronics design, backend, frontend, database architecture, embedded programming, machine learning, research).

ISS Interactive Simulation Experience (ISS app)

NASA Space Apps Hackathon

Built a React and Godot-based platform to visualize International Space Station data and recreate the ISS Cupola experience through interactive simulation and virtual image capture (frontend, simulation development, game design, API integration).

Skills

Programming

Backend & Software Development (JavaScript, PHP, C#, C/C++ , Python) | Web Development (HTML, CSS, .NET, Node.js, Laravel, React, Bootstrap) | Database Management (PostgreSQL, MySQL, Oracle Database, SQLite, MongoDB) | Data Analysis & Visualization (Pandas, NumPy, Matplotlib, Seaborn, Jupyter) | Machine Learning (Scikit-learn) | Embedded & IoT (ESP32, Arduino, Raspberry Pi, PLCs, PlatformIO, Node-RED) | Operating Systems (Ubuntu, Debian, macOS, Windows) | Containerization & Cloud (Docker, OCI) | Networking & Communication Protocols (Serial, I2C, SPI, Wi-Fi, ESPNow)

Infrastructure & Tools

Version Control (Git, GitHub) | API Development & Testing (Postman, Swagger)

Methodologies

Agile & Scrum | Development Practices (REST API Design, Clean Code, SOLID Principles, Git Flow)

Languages

Spanish (Native) | English (B2 - TOEFL)